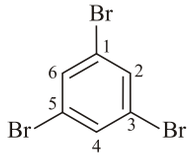
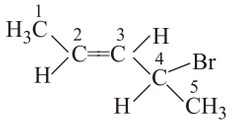
Halogen Derivatives

Exercise 1 : Basic level

1. (*b*) The IUPAC name is 1, 3, 5-tribromobenzene.



2. (*c*) The IUPAC name of the given compound is 4-bromopent-2-ene.



3. (*a*) Benzylic halides have a halogen attached to a benzyl carbon ($C_6H_5CH_2$—).

4. (*c*) Halogen derivatives contains at least one halogen atom. CH_3Cl (methyl chloride) has a chlorine atom, making it a halogen derivative.

5. (*c*) $\overset{2}{C}H_3—\overset{1}{C}H_2—Cl$ chloroethane

6. (*c*) Alkyl halides contains a halogen ($X = Cl,\ Br,\ I,\ F$) attached to an alkyl group.

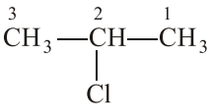
7. (*a*) The common name of CH_3I is methyl iodide while its IUPAC name is iodomethane.

8. (*a*) Aryl halides have a halogen directly attached to an aromatic ring C_6H_5Cl is an aryl halide.

9. (*c*) IUPAC name of C_6H_5Br, bromobenzene.

10. (*c*) CCl_4 (carbon tetrachloride) contains four chlorine atoms, making it a tetrahalide.

11. (*b*) The longest chain has 3 carbons.



2-chloropropane

12. (*b*) In *vicinal* dihalide, halogen atoms are attached to adjacent carbon atoms.

13. (*a*) The carbon bonded to Br is attached to only one other carbon.

14. (*b*) Benzyl halide has a halogen attached to a benzyl ($C_6H_5CH_2$) group.

15. (*d*) The boiling point of alkyl halides increases with molecular weight and stronger van der Waals' forces.

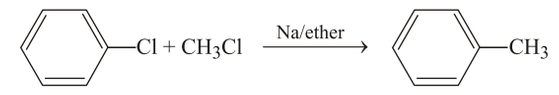
16. (*b*) The C—X bond is polar, creating strong dipole-dipole interaction raising the boiling point.

17. (*b*) Straight chain alkyl halides have greater boiling point than their isomers. This is due to the reason that with branching, the surface area of the alkyl halide decreases and hence the magnitude of the van der Waals' forces of attraction decreases. Therefore, $CH_3CH_2CH_2CH_2Cl$ has highest boiling point.

18. (*c*) Iodine is the heaviest halogen leading to stronger London dispersion forces and a higher boiling point.

19. (*b*) Alkyl halide dissolve well in organic solvent like ether due to non-polar interaction.

20. (*c*)



Chlorobenzene — Toluene

The above reaction is **Wurtz-Fittig reaction**.

21. (*c*) Enantiomers rotate plane-polarised light in opposite direction.

22. (*a*) Recemic mixture contain equal amount of both enantiomers canceling out optical rotation.

23. (*d*) Etheral solution of an alkyl halide gives an alkane when heated with metallic sodium. This is called **Wurtz reaction**.

$$2RX + 2Na \xrightarrow{\text{Dry ether}} R—R + 2NaX$$

24. (*a*) $RX + AgCN \longrightarrow \underset{\text{Alkyl isocyanide}}{RNC} + AgX$

When alkyl halide reacts with silver cyanide, isocyanides are obtained. It is due to nucleophilic substitution reaction.

25. (*b*) $\underset{\text{Ethyl iodide}}{2CH_3CH_2I} + Ag_2O \xrightarrow{\Delta} \underset{\text{Diethyl ether}}{(CH_3CH_2)_2O} + 2AgI$

26. (*a*) $\underset{\text{Alcohol}}{R\ OH} + PCl_5 \longrightarrow \underset{\text{Alkyl halide}}{R—Cl} + POCl_3 + HCl$

27. (*b*) Alkyl iodides are often prepared by the reaction of alkyl chlorides or bromides with NaI in dry acetone. This reaction is known as **Finkelstein reaction**.

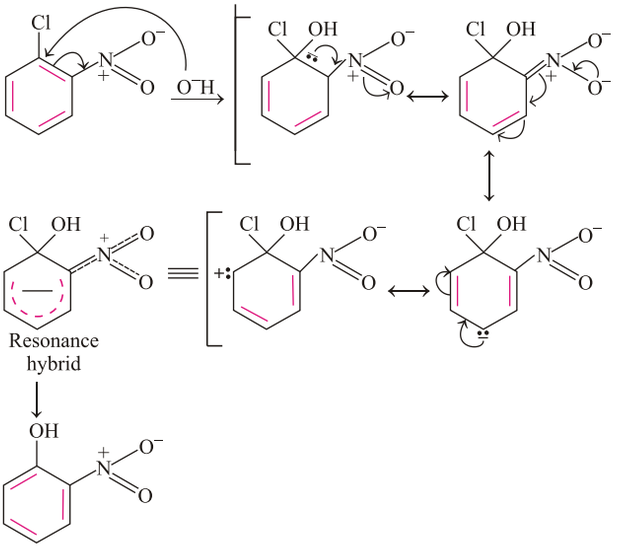
$$RX + NaI \longrightarrow RI + NaX$$

28. (*c*) Among the given reactions, Swarts reaction is

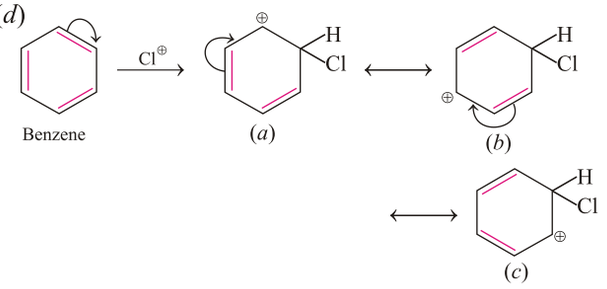
$$CH_3Br + AgF \longrightarrow CH_3F + AgBr$$

It is a type of halogen exchange reactions.

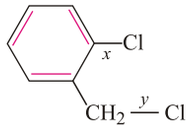
29. (*c*) The presence of electron withdrawing groups such as —NO_2, —CN etc. At *o*-and *p*-positions with respect to the halogen greatly activates the halogen towards nucleophilic displacement.



30. (*d*)



31. (*b*)



Due to involvement of '*x*' bond in resonance it acquires some double bond character hence its bond length is lesser than *y*.

32. (*c*) The correct stability order of alkyl substituted alkenes is $R_2C{=}CR_2 > R_2C{=}CHR > R_2C{=}CH_2$.

33. (*b*) Haloarenes undergo the electrophilic reactions of benzene ring such as halogenation, nitration, sulphonation and Friedel-Craft reactions.

34. (*b*) Thionyl chloride ($SOCl_2$) is used to convert alcohols into alkyl chloride, often giving better yields than other methods.

35. (*d*) Tertiary alcohols react faster with H*X* due to the formation of a stable tertiary carbocation.

36. (*d*) $CH_3{-}CH_2{-}OH \xrightarrow[\Delta]{HBr} \underset{\text{Bromoethane}}{CH_3CH_2{-}Br}$

37. (*b*) $ZnCl_2$ (Lucas reagent) helps activates HCl, making it more reactive towards primary and secondary alcohols.

38. (*a*) $SOCl_2$ converts alcohols to alkyl chlorides, producing HCl and SO_2 as gaseous by products.

$$R{-}OH + SOCl_2 \longrightarrow R{-}Cl + HCl + SO_2\,(g)$$

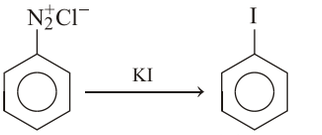
39. (*c*) Finkelstein reaction involves the exchange of Cl or Br with using NaI in acetone.

$$R{-}X \xrightarrow{NaI/\,Acetone} R{-}I + NaX$$

40. (*a*) $FeBr_3$ activates Br_2 making it more electrophilic for substitution on benzene.

41. (*b*) Sandmeyer's reaction convert aryl diazonium salt to aryl halide using CuCl, CuBr, CuCN.

42. (*c*)



43. (*b*) Swarts reaction replaces Br or Cl with F using AgF or SbF_3.

$$R{-}Br \xrightarrow{AgF} R{-}F + AgBr$$

44. (*c*) Phosphorus tribromide (PBr_3) is commonly used to convert alcohols to alkyl bromides.

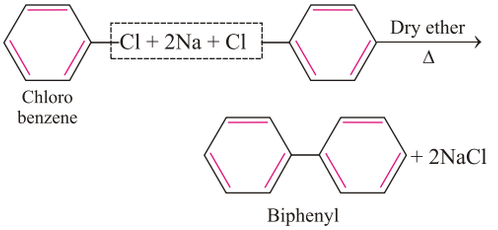
45. (*b*) Finkelstein reaction involves the exchange of Br / Cl with I using NaI in acetone.

46. (*a*) Hydrobromic acid (HBr) can be generated in *situ* by reacting sodium bromide (NaBr) with sulphuric acid. This method is commonly used to convert alcohols into alkylbromides.

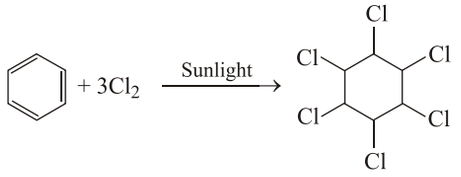
47. (*b*) Sandmeyer reaction uses CuBr to replace the diazonium group ($-N_2^+$) with Br.

48. (*b*) Sandmeyer reaction uses CuCl to replace the diazonium ($-N_2^+$) group with chlorine.

49. (*a*) Reaction of aryl halides with sodium in dry ether to form diaryl is known as **Fittig reaction**.



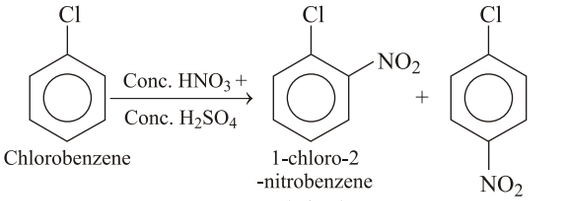
50. (*a*) Gammexane is chemically known as benzene hexachloride (BHC) though the latter name is incorrect but the older name (BHC) is still in use.



Gammexane (BHC)

51. (*c*) Halogen atom being slightly deactivating is *o*, *p*-directing.

52. (*c*)



Chlorobenzene | 1-chloro-2-nitrobenzene (minor) | 1-chloro-4-nitrobenzene (major)

53. (*c*) The E2 mechanism is favoured by strong bases like *t*-buOK.

54. (*b*) Grignard reagents (*R*—Mg*X*) are prepared using magnesium in dry ether.

55. (*a*) It is Wurtz reaction.

$$C_2H_5I + 2Na + C_2H_5I \xrightarrow{\text{Dry ether}} \underset{\text{Butane}}{C_2H_5 — C_2H_5} + 2NaI$$

So, C_4H_{10} (alkane) is formed.

56. (*d*) As $C_6H_5 — \overset{\overset{CH_3}{|}}{\underset{\underset{C_6H_5}{|}}{C^+}}$ carbocation is most stable.

Therefore, $C_6H_5C(CH_3)(C_6H_5)Br$ readily undergoes S_N1 reaction.

57. (*a*) $\underset{\text{Butane}}{CH_3CH_2CH_2CH_3} \xrightarrow[\text{UV light}]{Cl_2} \underset{\text{1-chlorobutane}}{CH_3CH_2CH_2CH_2Cl} + \underset{\text{2-chlorobutane}}{CH_3CH_2\underset{\underset{Cl}{|}}{CH}CH_3}$

58. (*b*) Statement (b) is incorrect about S_N2 mechanism.

It's correct form is as follows

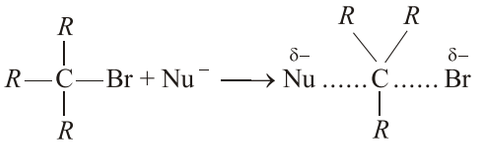
In transition state, all three C—H bonds are in the same plane and the attacking and leaving nucleophiles are partially attached to the carbon.

Rest other statements are correct.

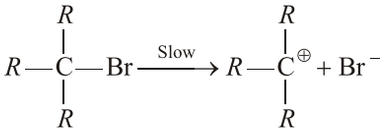
59. (*c*) —OH group is converted into —Cl group by $SOCl_2$ or anhydrous $ZnCl_2$/Conc. HCl or HCl etc.

$$\underset{\text{Alcohol}}{R — OH} \xrightarrow[\text{Anhyd. } ZnCl_2]{\text{Dry HCl}} \underset{\text{Alkyl halide}}{R — Cl} + H_2O$$

60. (*a*) In S_N2 reaction, nucleophile and alkyl halide react in one step.



Thus, tertiary carbon is under steric hindrance due to presence of three bulky alkyl groups thus reaction does not take place until (C—Br) bond breaks.



61. (*a*) Allyl halides are those compounds in which halogen atom is bonded to sp^3-hybridised carbon atom next to carbon-carbon double bond, i.e. to an allylic carbon.

62. (*d*) $ROH + HCl \xrightarrow{ZnCl_2} RCl + H_2O$

$$ROH + Br_2 \xrightarrow{\text{Red P}} R — Br$$

63. (*c*) $\underset{\text{Ethyl bromide}}{CH_3CH_2Br} + AgCN \longrightarrow \underset{\text{Ethyl isocyanide}}{CH_3CH_2NC} + AgBr$

64. (*c*) Clearly, rate of S_N1 reaction depends on leaving group tendency (— I > — Br > — Cl > — F) as well as on stability of carbocation formed (3° > 2° > 1°).

65. (*a*) Wurtz-Fittig reaction is the reaction of unsymmetrical alkyl halide.

$$C_6H_5I + 2Na + CH_3I \xrightarrow[\text{Dry acetone}]{NaI} \underset{\text{Toluene}}{C_6H_5CH_3} + 2NaI$$

66. (*c*) The given reaction follows S_N2 mechanism and these reactions are favoured in polar aprotic medium, like DMSO, DMF... etc.

Complete reaction is as follows

$$\underset{\text{1-bromopropane}}{CH_3CH_2CH_2Br} + NaCN \xrightarrow{DMF} \underset{\text{Butanenitrile}}{CH_3CH_2CH_2CN} + NaBr$$

67. (*d*) Alkyl halide are very slightly soluble in water because they do not form H-bonds with water. Fluoro and chloroalkanes are lighter than water while bromo, iodo are heavier than water.

Boiling point of alkyl halides are higher than corresponding alkane.

68. (*b*) The byproduct (*A*) formed in the reaction is

Complete the reaction is

$$\underset{\text{Alcohol}}{ROH} + PCl_3 \longrightarrow R\text{—}Cl + \underbrace{H_3PO_3}_{(A)}$$

69. (*a*) When haloalkane containing β-hydrogen atom is heated with alc. KOH solution, then alkene is formed as a result of elimination.

Reaction involved is as follows

$$\underset{\text{1-bromopropane}}{CH_3CH_2CH_2Br} \xrightarrow{\text{Alc. KOH}} \underset{\text{Propene}}{CH_3CH=CH_2}$$

70. (*c*) The S_N2 mechanism is a bimolecular reaction where the rate depend on both reactants.

71. (*c*) Elimination reaction can proceed *via* E1 or E2.

72. (*c*) Alcoholic KOH favours elimination while aqueous KOH favours substitution.

73. (*c*) S_N2 reaction are favored in polar a protic solvent like acetone and DMSO, which do not stabilize carbocation.

74. (*c*) S_N1 Reaction rate depend on the formation of a stable carbocation, tertiary carbocation are the most stable.

75. (*a*) Alkyl halides react with magnesium in dry ether to form Grignard reagents ($R = MgX$)

76. (*c*) The wurtz reaction is used to couple alkyl halide using sodium in dry ether.

77. (*d*) Aryl halide do not undergo S_N1 reaction due to the instability of the aryl carbocation.

78. (*c*) S_N1 reaction proceed fastest with tertiary alkyl halides due to carbocation stability.

79. (*c*) Two ethyl radicals couple to form butane in the Wurtz reaction.

80. (*c*) The Wurtz Fitttig reaction couples alkyl halides with aryl-halides using sodium in dry ether.

81. (*a*) Sodium metal facilitates coupling of alkyl and aryl halides.

82. (*b*) Ethanol (—OH) is not a halogen derivatives.

83. (*b*) Sulphonation involves electrophilic substitution using concentrated sulphuric acid.

84. (*d*) DDT (Dichloro Diphenyl Trichloro Ethane) is a powerful insecticide but is not easily biodegradable. Therefore, its use is banned in many countries.

85. (*c*) CCl_4 at higher temperature reacts with water to give phosgene.

$$CCl_4 + H_2O \xrightarrow{500^\circ C} \underset{\text{(Phosgene)}}{COCl_2} + 2HCl$$

Phosgene is highly poisonous, so after the use of pyrene, CCl_4 to extinguish fire, the room should be well ventilated.

86. (*a*) Dichloromethane can be metabolized to carbon monoxide, causing liver damage.

87. (*a*) Dichloromethane is widely used as an industrial solvent and in paint removers.

88. (*a*) Methyl chloride is a toxic and can affect the nervous system and respiratory function.

89. (*a*) Chloroform was used as an early anesthetic but was discontinued due to toxicity.

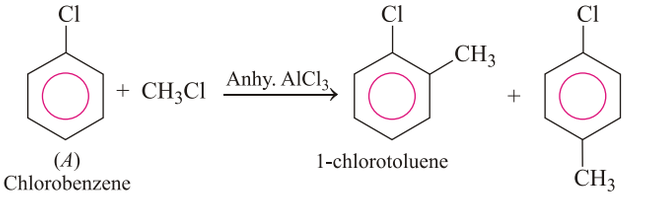
90. (*a*) Chloroform is metabolized in the liver and can cause liver damage..

Halogen Derivatives

Exercise 3 : PYQ Section

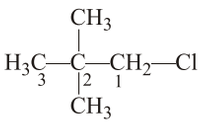
1. (*b*) Chlorobenzene reacts with chloromethane in the presence of anhydrous $AlCl_3$ to give 1-chlorotoluene and 4-chlorotoluene.

The reaction involved is as follows



(*A*) Chlorobenzene — 1-chlorotoluene — 4-chlorotoluene

2. (*a*)



The IUPAC name of neopentyl chloride is 1-chloro-2-dimethyl propane.

3. (*a*) BHT (Butylated hydroxytoluene) is used as an antioxidant.

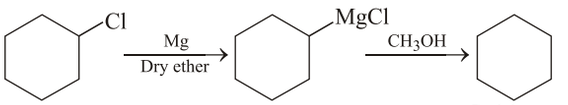
4. (*a*) Silver fluoride acts as a fluorinating agent, replacing the bromide ion with fluoride to form *tert*-butyl fluoride.

5. (*c*) Williamson's synthesis involves the reaction of primary alkyl halide with alkoxides to form ethers.

6. (*c*) Among the given compounds, 1-bromo butane, being the longest straight chain molecule, offers better packing than branched isomers and thus has a higher melting point.

7. (*b*) $CaCO_3$ in crystalline form is used prepare Nicol prism.

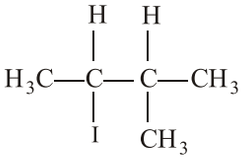
8. (*d*)



Cyclohexane (*B*)

9. (*b*) $Ar—X + R—X + 2Na \xrightarrow{\text{Dry ether}} Ar—R + 2NaX$

10. (*c*) A chiral carbon must be attached to four different group atoms.

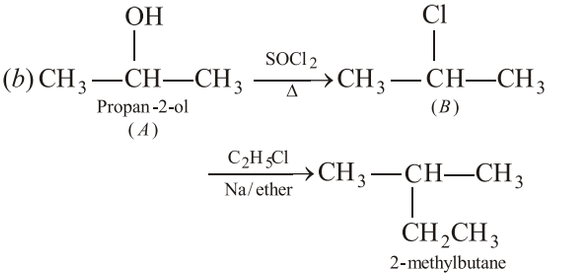


2-iodo-3-methylbutane.

11. (*d*) Greater the number of electron withdrawing groups ($—NO_2$) at *o/p*-position, greater is the reactivity.

2, 4, 6-trinitrochlorobenzene has the highest reactivity towards nucleophilic substitution reaction.

12. (*b*)

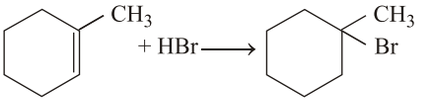


Propan-2-ol (*A*) — (*B*) — 2-methylbutane

13. (*c*) $R—X + 2Na + Ar—X \longrightarrow R—Ar + 2NaX$, is Wurtz Fittig reaction.

14. (*b*) Vinyl halide are a type of organic compound where a halogen atom is directly bonded to carbon-carbon double bond.

15. (*c*)



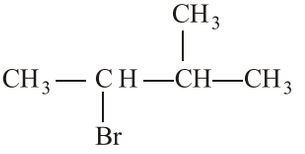
1-bromo-1methyl cyclohexane

Product formed according to Markownikoff rule.

16. (*c*) When ethyl chloride react with alc.KCN ethyl cyanide is formed.

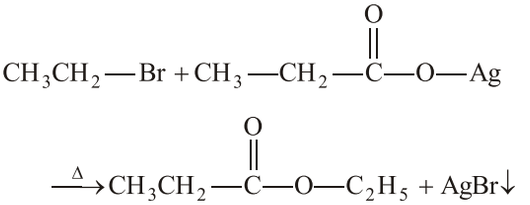
$$\underset{\text{Ethyl chloride}}{C_2H_5Cl} + \text{Alc.KCN} \xrightarrow{\Delta} \underset{\text{Ethyl cyanide}}{C_2H_5CN} + KCl$$

17. (*c*) A chiral molecule is a molecule that cannot be superimposed on its mirror image, due to presence of an asymmetric carbon atom



2-bromo-3-methylbutane

18. (*c*) Silver propanoate is used in this conversion

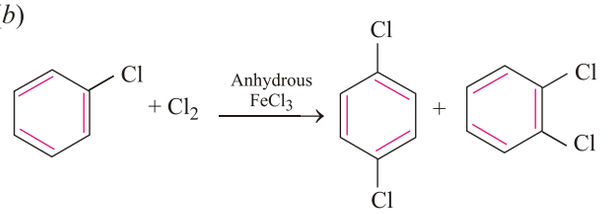


19. (*d*) $CH_2 = CH—CH_3 + Br_2 \longrightarrow$

$$CH_2 = CH—CH_2Br + HBr$$

20. (*c*) $CHCl_3 \xrightarrow{[O]} \underset{\text{Phosgene gas}}{COCl_2} + 2HCl$

21. (*b*)



Major
1,4-dichlorobenzene

22. (*d*) Fittig reaction involves reaction of aryl halide with sodium metal.

23. (*a*) Straight chain compound have high van der Waals' force and the surface area is more thus high boiling point.

24. (*a*) In haloalkyne, halogen is attached to alkyne.

$$CH \equiv C—Cl$$

25. (*d*) $nCH_3MgI + H_2O \xrightarrow{\text{Dry ether}} nCH_4 + Mg(OH)I$

26. (*d*) $\underset{\text{Alkyl halide}}{R—X} \xrightarrow{KNO_2} \underset{\text{Alkyl nitrite}}{R—O—N = O}$

27. (*b*) In Wurtz reaction product will be

$$CH_3Cl + CH_3CH_2CH_2Cl \xrightarrow[\text{Dry ether}]{Na} CH_3—CH_3$$

$$+ CH_3—CH_2—CH_2—CH_3$$
$$+ CH_3—CH_2—CH_2—CH_2—CH_2—CH_3$$

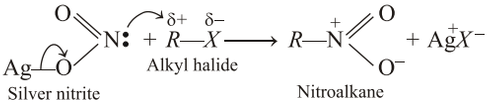
28. (*b*) The IUPAC name of the given compound is 4-bromo-4 methylpent-2-ene.

29. (*c*) Grignard reagent being basic in nature will accept H^+ from NH_3 and forms methane and H_2N MgBr.

The reaction is represented as

$$CH_3Mg\ Br \xrightarrow{NH_3} CH_4 + H_2NMgBr$$

30. (*d*) The reaction is represented as



Silver nitrite Alkyl halide Nitroalkane

31. (*b*) Coupling of aryl halide with alkyl halide in presence of sodium metal in dry ether known as Wurtz-Fittig reaction.

32. (*c*) Bond enthalpy of $CH_3—F$ moleule is 452 kJ mol^{-1}.

33. (*d*) $R—X + Mg \xrightarrow{\text{Dry ether}} \underset{\text{Grignard reagent}}{R—Mg—X}$

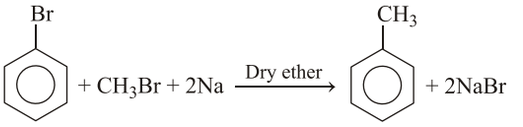
34. (*b*) The bond length between carbon and bromine in $CH_3—Br$ is 193 pm.

35. (*a*) Reactivity of C—*X* bond $\propto$ no. of EWG at *o* and *p*-position.

36. (*c*) Order of reactivity of haloacid 3° > 2° > 1°.

37. (*b*) Tertiary alkyl halide prefer S_N1 mechanism.

38. (*d*) Wurtz-Fittig reaction is used to prepare toluene from bromobenzene. The reaction involve is as follows



Bromobenzene Toluene

39. (*a*)

$$CH_3CH_2—OH \xrightarrow[\Delta]{SOCl_2} \underset{X}{CH_3—CH_2—Cl} \xrightarrow[\text{Dry ether}]{Mg} \underset{(Y)}{CH_3CH_2MgCl}$$

$$\xrightarrow{NH_3} \underset{\substack{(Z)\\\text{Ethane}}}{CH_3—CH_3} + Mg\begin{matrix}Cl\\NH_2\end{matrix}$$

40. (*c*) Boiling point increases with increase in molecular weight.

41. (*d*) Benzylic halide is an organic compound in which halogen atom is directly bonded to a carbon atom that is adjacent to an aromatic benzene ring.

42. (*c*) Rate of nitration is faster when substituent activates the ring (+*I*-effect or +*R*-effect, *ortho/para*-directing) and rate is slower when substituent deactivates the ring (–*I*-effect or –*R*-effect, *meta*-directing group).

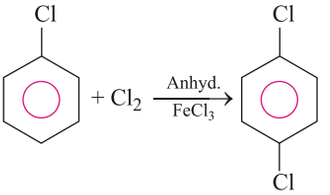
- Halogen deactivates ring but orientation is *ortho-para* directing group.
- Rate of $C_6H_6 = C_6D_6$, as no effect is observed when H is replaced by D. Hence, order is I = II > IV > III.

43. (*d*) In the Wurtz reaction, alkyl halides react with sodium in dry ether to give hydrocarbon containing double the number of carbon atoms present in the halide.

$$\underset{\substack{\textit{tert}\text{-butyl bromide}}}{2(CH_3)_3CBr} \xrightarrow{Na/\text{Dry ether}}$$

$$\underset{\substack{2,2,3,3\text{-tetramethyl butane}}}{H_3C—C(CH_3)_2—C(CH_3)_2—CH_3} + \underset{\substack{\text{Sodium bromide}}}{2NaBr}$$

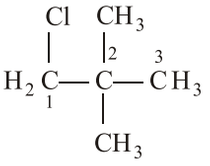
44. (*c*) Cl_2 in presence of $FeCl_3$ attack on benzene ring. So, given rection is



1,4-dichlorbenzene
(Major)

45. (*c*) The correct order of hydrogen halides is HI > HBr > HCl. The cleavage of ethers usually takes place with conc. HI or HBr at high temperature. Greater the nucleophilicity of halide ion, more is the reactivity of hydrogen halide.

46. (*b*) *Neo*-pentyl chloride is common name of 1-chloro-2, 2-dimethyl propane.



1-chloro-2, 2-dimethyl propane (*neo*-pentyl chloride)

47. (*d*) Chlorination of alkanes in presence of sunlight is not suitable for the preparation of alkyl chloride because chlorination of alkanes gives a complex mixture of isomeric mono and polyhaloalkanes, which is difficult to separate as pure compounds. Consequently, the yield of any one compound is low.

$$\underset{\text{Butane}}{CH_3CH_2CH_2CH_3} \xrightarrow{Cl_2/UV\ light} CH_3CH_2CH_2CH_2Cl + CH_3CH_2CH(Cl)CH_3$$

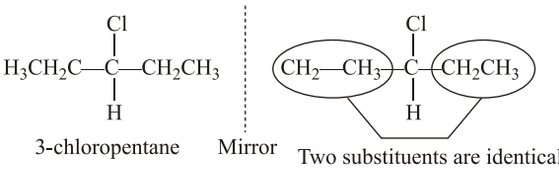
48. (*a*) As the size of halogen atom (X) increases, the C—X bond length increases and hence, the C—X bond strength decreases. Since, the order of size of halogen atom is, F < Cl < Br < I

Thus, the order of C—X bond strength is,

$$C—F > C—Cl > C—Br > C—I$$

The correct order of C—X bond strength in CH_3—X is, $CH_3F > CH_3Cl > CH_3Br > CH_3I$

49. (*b*) 3-chloropentane compound is not optically active.



Because, in this compound no stereogenic centre is present and they are superimposable on each other. So, in the molecule absence of stereogentic centre shows that this molecule is not optically active.

50. (*d*) More stable is carbocation formed, more $\overset{\oplus}{}$the reactivity towards S_N1 mechanism CH_3—Br gives CH_3 which is not stable, hence least reactive S_N1 mechanism.

51. (*d*) Aryl iodide and aryl fluoride cannot be prepared directly by electrophilic substitution because reaction with iodine are reversible in nature and require the presence of an oxidising agent (HNO_3, HIO_4) to oxidise the HI formed.

Fluoro compounds are not prepared by this method due to high reactivity of fluorine.

Aryl iodide + aryl fluoride ⟶ no reaction

52. (*d*) $CH_3CH_2Br + 2[H] \xrightarrow[\text{Alcohol}]{\text{Zn - Cu}} CH_3—CH_3 + HBr$

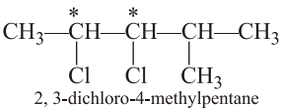
[X] is Zn -Cu couple in alcohol. In the reaction, when ethyl bromide is reacted with Zn -Cu couple and alcohol, ethyl bromide reduced to give ethane. This is more or less a redox reaction. In it, ethyl bromide is reduced to give an ethyl organo metallic and Br^- subsequently, the ethanol protonates the ethyl organo metallic to give ethane.

53. (*c*) CF_2Cl_2 is used as refrigerants.

54. (*a*) No. of optical isomer possible $= 2^n$, where n is no. of chiral carbon atom.

$\because$ $n = 3, 2^3 = 8$

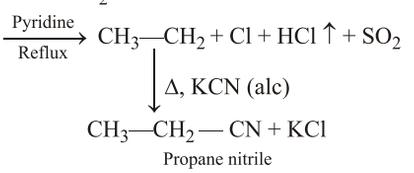
55. (*c*) There are two asymmetric carbon atom



2, 3-dichloro-4-methylpentane

56. (*c*) Halogen attached directly aromatic compound is classified as aryl halide.

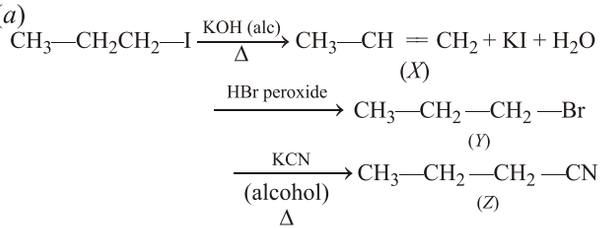
57. (*a*)



58. (*d*) Freons are used as foaming agent in foamed plastic manufacturing.

59. (*d*) Swarts reaction is an example of halogen exchange method for preparation of halide derivative.

60. (*a*)



61. (*d*) Structural formula of allyl chloride

$$CH_2 = CH—CH_2—Cl$$

Molecular formula will be C_3H_5Cl.

62. (*c*) Higher the molecular weight of halide more will be the boiling point.

63. (*c*) When propene reacts with HCl in presence of peroxide, the product obtained is 2-chloropropane. This reaction in proceeds according to Markownikoff's rule because some of the steps in chain reaction are endothermic nature.

The reaction is as follow

$$H_2C{=}CH{-}CH_3 + HCl \xrightarrow{\text{Peroxide}} \underset{\text{2-chloropropane}}{CH_3{-}\overset{Cl}{\overset{|}{C}}H{-}CH_3}$$

(Propene)

64. (*a*)

$$H_3C{-}\overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}{-}CH_3 \xrightarrow[h\nu]{X_2} H_3C{-}\overset{CH_3}{\overset{|}{\underset{CH_3}{\underset{|}{C}}}}{-}CH_2X + HX$$

(*neo*-pentane)

As it contains only one type of hydrogen, so one type of monohalogen derivative is formed.

65. (*d*) $C_2H_5MgBr + H_2O \longrightarrow CH_3{-}CH_3 + Mg(OH)Br$

66. (*b*) $R{-}Cl + NaI \xrightarrow{\text{Acetone}} R{-}I + NaCl$ (Finkelstein reaction).

67. (*d*) The Wurtz reaction between methyl chloride and sodium metal in presence of dry ether can be written as

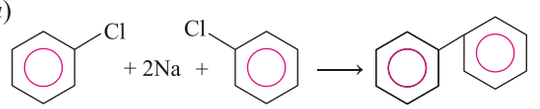
$$2CH_3Cl + 2Na \xrightarrow{\text{Dry ether}} \underset{\text{Ethane}}{CH_3{-}CH_3} + 2NaCl$$

2 moles of sodium metal reacts to give 1 mole of ethane also, 1 mole of Na metal = 23 g

$\therefore$ 2 moles of Na metal = $23 \times 2 = 46$ g

Thus, 46 g of sodium is required to prepare one mole of ethane from methyl chloride by Wurtz reaction.

68. (*a*)



69. (*a*) Alkaline hydrolysis of 1-bromo-1-phenylethane produce 1-phenylethan-1-ol. The reaction proceed *via* S_N1 reaction. A carbocation is formed in the slow step of S_N1 reaction.

The positively charged carbon of the carbocation intermediate is sp^2- hybridised and the three bonds connected to an sp^2- hybridised are in the same plane.

Reaction involved is as follows

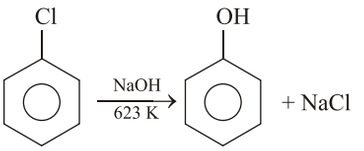
$$\underset{\substack{\text{1-bromo}\\\text{-1-phenyl ethane}}}{CH_3{-}\overset{Br}{\overset{|}{C}}H{-}Ph} \rightleftharpoons \underset{sp^2\text{ hybridised}}{CH_3{-}\overset{\oplus}{C}H{-}Ph} \xrightarrow[S_N1\text{ reaction}]{OH^{\ominus}} CH_3{-}\overset{OH}{\overset{|}{C}}H{-}Ph$$

70. (*d*) $2nR{-}X \xrightarrow[\text{Dry ether}]{+\,2n\text{Na}} \underset{\text{Alkane}}{R{-}R} + 2\text{Na}X$

The above given reaction is known as **Wurtz reaction**.

71. (*c*) Methylene dichloride is used as a solvent in the direct method of decaffeination of the coffee.

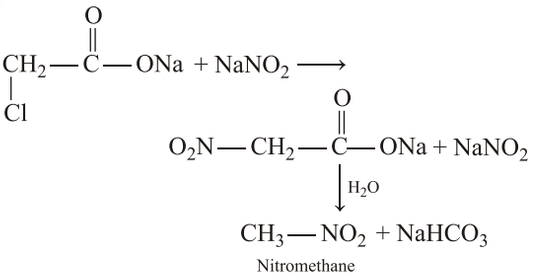
72. (*c*) (Nucleophilic substitution reaction).



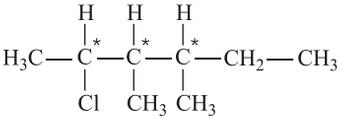
73. (*a*) In Laboratory, nitro alkanes are produced by reaction of 1° and 2° alkyl halides with silver nitrite. This is example of nucleophilic substitution reaction.

$$R{-}Br/RI + AgNO_2 \xrightarrow[\text{Alcohol}]{\Delta} R - NO_2 + AgBr/AgI$$

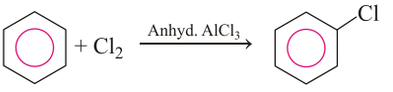
74. (*a*)



75. (*c*)

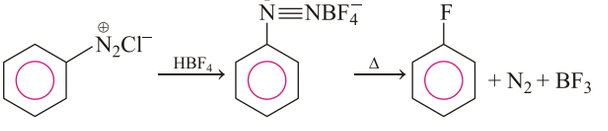


76. (*a*) Electrophilic substitution reaction



77. (*a*) Reaction of alkanes with fluorine are explosive. This is an explosive process which is difficult to control, so carried out in dark. In this process, replacement of one or more hydrogen of alkane by halogen takes place. For this reaction, the order of reactivity of halogen is $I_2 < Br_2 < Cl_2 < F_2$.

78. (*b*) Balz-Schemann reaction

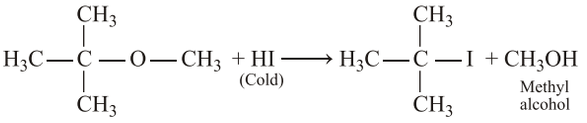


79. (*d*) Enantiomer have different value in their specific rotation.

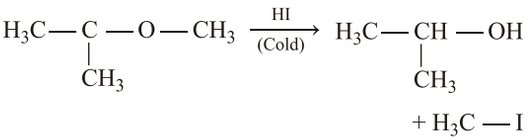
80. (*b*) $CH_3{-}CH_2{-}Br + NaI \xrightarrow{\text{Dry acetone}} CH_3{-}CH_2{-}I + NaBr$

This is Finkelstein reaction.

81. (*b*) The reaction proceeds as

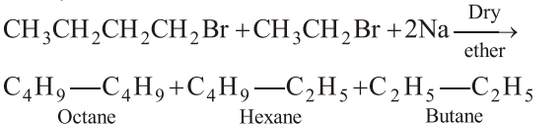


82. (*b*)



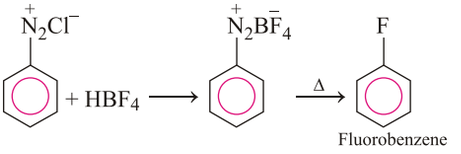
83. (*d*) In Wurtz synthesis, the product formed always contain more number of carbon atom than reactant.

Thus, ethane cannot be formed.

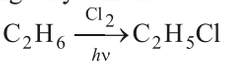


84. (*d*) Fluoride smallest in size and highest electronegativity, so magnesium fluoride has highest ionic character.

85. (*c*) When diazonium salt is treated with fluoroboric acid (HBF_4), benzene diazonium fluoroborate is precipitated, which on heating gives fluorobenzene. This reaction is called **Balz-Schiemann reaction**.



86. (*c*) Chlorination of ethane is carried out in the presence of ultraviolet-light resulting free radical reaction forming ethyl chloride.



87. (*c*) Chemical composition of Nicol's prism is $CaCO_3$.

88. (*d*)

